

Outline of the lecture

- MINI IZATION TECHNIQUES
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 - HOW DO WE DO IT
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 - K-MAP
 - QM METHOD
- BOOLEAN ALGEBRA
 - AXIOMS OF BOOLEAN ALGEBRA
 - LAWS OF BOOLEAN ALGEBRA
 - ✓ Complementation Laws
 - ✓ AND Laws, OR Laws
 - ✓ Commutative Laws, Associative Laws
 - ✓ Distributive Laws
 - ✓ Idempotence Laws
 - ✓ Negation Laws, Double Negation Laws
 - ✓ Identity Laws, Null Laws
 - ✓ Absorption Laws
 - ✓ Consensus Theorem
 - ✓ Transposition Theorem
 - ✓ De Morgan's Theorem
 - DUALITY
- EXAMPLES
- PROBLEMS FOR PRACTICE

MINIMIZATION TECHNIQUES

- ▶ Realization of any Digital circuit requires gates
- ▶ Every logic operation represents an element of hardware. Reducing the expression to minimal, results in reduction of cost and complexity
- ▶ For the design to be efficient, minimum number of gates should be used in the circuit
- ▶ The circuit/equation/output function can be reduced or minimized using:
 - ▶ Boolean Algebra
 - ▶ Method involving Laws, Theorems & Axioms
 - ▶ Karnaugh - map (K-map)
 - ▶ Method of graphical representation
 - ▶ Quine - McClusky (QM) method
 - ▶ Tabulation method of minimization

BOOLEAN ALGEBRA

- ▶ It's a system of mathematical logic
- ▶ Complex logic statements can be expressed by Boolean function
- ▶ Comprises of well - developed set of Rules and Laws
- ▶ A variable in Boolean function has only logical value. It does not have any numerical significance.
- ▶ Can be represented as logic 1 or logic 0. Generally,
 - ▶ Logic 1 refers TRUE, ON, HIGH
 - ▶ Logic 0 refers False, OFF, LOW

BOOLEAN ALGEBRA- AXIOMS

- ▶ Set of logical expressions accepted without proof
- ▶ Axioms are based upon the operation performed by the basic gates i.e. AND, OR, NOT

Axiom 1	$0 \cdot 0 = 0$
Axiom 2	$0 \cdot 1 = 0$
Axiom 3	$1 \cdot 0 = 0$
Axiom 4	$1 \cdot 1 = 1$
Axiom 5	$0 + 0 = 0$
Axiom 6	$0 + 1 = 1$
Axiom 7	$1 + 0 = 1$
Axiom 8	$1 + 1 = 1$
Axiom 9	$\overline{1} = 0$
Axiom 10	$\overline{0} = 1$

BOOLEAN ALGEBRA- LAWS

- Complementation Laws : Means to INVERT. Five laws are:

Law 1	$\bar{1} = 0$
Law 2	$\bar{0} = 1$
Law 3	If $A = 0$, then $\bar{A} = 1$
Law 4	If $A = 1$, then $\bar{A} = 0$
Law 5	$\bar{\bar{A}} = A$

- AND Laws : Four laws are:

Law 1	$A \cdot 0 = 0$
Law 2	$A \cdot 1 = A$
Law 3	$A \cdot A = A$
Law 4	$A \cdot \bar{A} = 0$

- OR Laws : Four laws are:

Law 1	$A + 0 = A$
Law 2	$A + 1 = 1$
Law 3	$A + A = A$
Law 4	$A + \bar{A} = 1$

BOOLEAN ALGEBRA- LAWS

- Commutative Laws : Allows change in position of AND/OR variables

Law 1	$A + B = B + A$ $A + B + C = B + A + C$ $= C + B + A$
Law 2	$A \cdot B = B \cdot A$ $A \cdot B \cdot C = B \cdot A \cdot C$ $= C \cdot B \cdot A$

- Associative Laws : Allows grouping of variables

Law 1	$(A + B) + C = A + (B + C)$ $(A + B + C) + D = A + (B + C + D)$ $= (A + B) + (C + D)$
Law 2	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$ $(A \cdot B \cdot C) \cdot D = A \cdot (B \cdot C \cdot D)$

BOOLEAN ALGEBRA- LAWS

- **Distributive Laws** : Allows factoring or multiplying out of expression. Three laws are:

Law 1	$A(B + C) = A.B + A.C$ $A.B (C +D) = A.B.C + A.B.D$
Law 2	$A + B.C = (A + B)(A + C)$ <u>PROOF</u> : $(A + B)(A + C)$ $= A.A + A.C + B.A + B.C$ $= A + A.C + A.B + B.C$ $= A(1 + C + B) + B.C$ {1+C=1;OR law} $= A + B.C$
Law 3	$A + \bar{A} . B = A + B$ <u>PROOF</u> : $A + \bar{A} . B$ $= (A + \bar{A}) . (A + B)$ $= 1. (A + B)$ $= A + B$

- **Idempotence Laws** : Idempotence means same value

Law 1	$A + A = A$
Law 2	$A . A = A$

BOOLEAN ALGEBRA- LAWS

- Negation Laws : states ANDing or ORing of the variable with its complement

Law 1	$A \cdot \bar{A} = 0$
Law 2	$A + \bar{A} = 1$

- Double Negation Laws

Law 1	$\bar{\bar{A}} = A$ **Odd number of inversions $\equiv$ Single inversion Even number of inversions $\equiv$ No inversion
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- Identity Laws

Law 1	$A \cdot 1 = A$
Law 2	$A + 1 = 1$

- Null Laws

Law 1	$A \cdot 0 = 0$
Law 2	$A + 0 = A$

BOOLEAN ALGEBRA- LAWS

► Absorption Laws

Law 1	$A + A \cdot B = A$ <u>Proof:</u> $A + A \cdot B$ $= A (1 + B)$ $= A \cdot 1 = A$ ** $A + A \cdot \text{ANY TERM} = A$
Law 2	$A \cdot (A + B) = A$ <u>Proof:</u> $A \cdot (A + B)$ $= A \cdot A + A \cdot B$ $= A + A \cdot B$ $= A (1 + B)$ $= A \cdot 1 = A$ ** $A \cdot (A + \text{ANY TERM}) = A$

BOOLEAN ALGEBRA- THEOREMS

► Consensus Theorem

Theorem 1	$A.B + \bar{A}.C + B.C = A.B + \bar{A}.C$ <u>Proof (LHS)</u> : $A.B + \bar{A}.C + B.C$ $= A.B + \bar{A}.C + B.C (A + \bar{A})$ $= A.B + \bar{A}.C + B.C.A + B.C . \bar{A}$ $= A .B (1+C) + \bar{A}.C(1 + B)$ $= A .B.1 + \bar{A}.C.1$ $= A .B + \bar{A}.C$ ** Can be extended to any number of variables ** $A.B + \bar{A}.C + B.C.D = A.B + \bar{A}.C$

BOOLEAN ALGEBRA- THEOREMS

► Consensus Theorem (cont.)

Theorem 2	$(A+B)(\bar{A}+C)(B+C) = (A+B)(\bar{A}+C)$ <u>Proof (LHS)</u> : $(A+B)(\bar{A}+C)(B+C)$ = $(A.\bar{A} + A.C + B.\bar{A} + B.C)(B + C)$ = $(A.C + B.C + \bar{A}.B)(B + C)$ = $A.B.C + B.C + \bar{A}.B + A.C + B.C + \bar{A}.B.C$ = $A.C + B.C + \bar{A}.B$ <u>Proof (RHS)</u> : $(A+B)(\bar{A}+C)$ = $(A.\bar{A} + A.C + B.C + \bar{A}.B)$ = $A.C + B.C + \bar{A}.B$ = LHS ** Can be extended to any number of variables ** $(A+B)(\bar{A}+C)(B+C + D) = (A+B)(\bar{A}+C)$

BOOLEAN ALGEBRA- THEOREMS

► Transposition Theorem

Theorem 1	$A.B + \bar{A}.C = (A+C)(\bar{A}+B)$ <u>Proof (RHS)</u> : $(A+C)(\bar{A}+B)$ $\begin{aligned} &= A.\bar{A} + C.\bar{A} + A.B + C.B \\ &= 0 + \bar{A}.C + A.B + B.C \\ &= \bar{A}.C + A.B + B.C(A + \bar{A}) \\ &= A.B + A.B.C + \bar{A}.C + \bar{A}.B.C \\ &= A.B(1 + C) + \bar{A}.C(1+B) \\ &= A.B + \bar{A}.C \\ &= \text{LHS} \end{aligned}$

BOOLEAN ALGEBRA- THEOREMS

► De Morgan's Theorem

Theorem 1	$\overline{A + B} = \bar{A} . \bar{B}$ ** Can be extended to any number of variables or combination of variables $\overline{A + B + C + D + \dots} = \bar{A} . \bar{B} . \bar{C} . \bar{D} . \dots$ $\overline{AB + CD + EFG + \dots} = \bar{A}\bar{B} . \bar{C}\bar{D} . \bar{E}\bar{F}\bar{G} . \dots$
Theorem 2	$\overline{A . B} = \bar{A} + \bar{B}$ ** Can be extended to any number of variables or combination of variables $\overline{A . B . C . D . \dots} = \bar{A} + \bar{B} + \bar{C} + \bar{D} + \dots$ $\overline{(AB) . (CD) . (EFG) . \dots} = \bar{A}\bar{B} + \bar{C}\bar{D} + \bar{E}\bar{F}\bar{G} + \dots$

BOOLEAN ALGEBRA- DUALITY

- ▶ Positive logic system and Negative logic system give rise to basic DUALITY in all Boolean identities
- ▶ When changing from one logic system to another
 - ▶ '0' becomes '1' and '1' becomes '0'
 $0 \Leftrightarrow 1$
 - ▶ AND gate becomes OR gate and OR gate becomes AND gate
 $\text{AND} \Leftrightarrow \text{OR}$
- ▶ For a given Boolean expression, to find a dual identity
 - ▶ Change all '+' signs to '.' signs
 - ▶ Change all '.' signs to '+' signs
 - ▶ Complement all 0s and 1s
 - ▶ Variables are not to be complemented

BOOLEAN ALGEBRA- DUALITY

► Few DUAL Identities are:

S. NO	EXPRESSION	DUAL
1.	$\bar{0} = 1$	$\bar{1} = 0$
2.	$0 \cdot 1 = 0$	$1 + 0 = 1$
3.	$0 \cdot 0 = 0$	$1 + 1 = 1$
4.	$1 \cdot 1 = 1$	$0 + 0 = 0$
5.	$A \cdot 0 = 0$	$A + 1 = 1$
6.	$A \cdot 1 = A$	$A + 0 = A$
7.	$A \cdot A = A$	$A + A = A$
8.	$A \cdot \bar{A} = 0$	$A + \bar{A} = 1$
9.	$A \cdot B = B \cdot A$	$A + B = B + A$
10.	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$	$A + (B + C) = (A + B) + C$
11.	$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + (B \cdot C) = (A + B)(A + C)$
12.	$A \cdot (A + B) = A$	$A + A \cdot B = A$
13.	$A \cdot (A \cdot B) = A \cdot B$	$A + A + B = A + B$
14.	$\overline{A \cdot B} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A} \cdot \bar{B}$

REDUCING BOOLEAN EXPRESSION

► The steps to reduce Boolean expressions using laws of Boolean algebra are

1. Expand the expression to remove parentheses

2. Look for identical terms $\{A+A = A\}$

3. Look for variable and its negation in same term
 $\{A.\bar{A}.B.C = 0\}$

4. Look for pairs of identical terms that differ by only one variable. The larger term is hence dropped
 $\{ABCD + ABC = ABC(D+1) = ABC\}$

5. Look for pairs of terms with one or more variables complemented. These variables are dropped and pairs are combined to single term
 $\{ABCD + \bar{A} BCD = BCD (A + \bar{A}) = BCD.1\}$

BOOLEAN ALGEBRA- THEOREMS

- ▶ De Morgan's Theorem
- ▶ Complicated expressions can be transformed by
 1. Complement the entire function
 2. Change all ANDs to ORs and all ORs to ANDs
 3. Complement each of the variables

Eg. Solve using DeMorgan's theorem

$$\overline{(A + \bar{B})(C + \bar{D})}$$

$$= \overline{(A + \bar{B})(C + \bar{D})}$$

$$= A.\bar{B} + C.\bar{D}$$

$$= \bar{A}.B + \bar{C}.D$$

Complementing the function

Change all ORs and all ANDs

Complement each variable

BOOLEAN ALGEBRA

EXAMPLES

Q1. Demorganize $\overline{\overline{AB}(CD + \overline{EF})(\overline{AB} + \overline{CD})}$

$$\begin{aligned} & \overline{\overline{AB}(CD + \overline{EF})(\overline{AB} + \overline{CD})} \\ &= \overline{\overline{AB}} + \overline{CD + \overline{EF}} + \overline{(\overline{AB} + \overline{CD})} \\ &= AB + (\overline{CD}.\overline{\overline{EF}}) + \overline{\overline{AB}}.\overline{\overline{CD}} \\ &= AB + (\overline{C} + \overline{D})(E + \overline{F}) + ABCD \end{aligned}$$

Q2. Reduce the expression $\overline{\overline{AB} + \overline{A} + AB}$

$$\begin{aligned} & \overline{\overline{AB} + \overline{A} + AB} \\ &= \overline{\overline{AB}.\overline{A}.AB} \\ &= \overline{AB.A.\overline{AB}} \\ &= \overline{AB.AB} \\ &= 0 \end{aligned}$$

BOOLEAN ALGEBRA

Q3. Prove that $\overline{A}BC + B + B\overline{D} + AB\overline{D} + \overline{A}C = B + C$

$$\begin{aligned}\text{LHS} &= \overline{A}BC + B + B\overline{D} + AB\overline{D} + \overline{A}C \\&= \overline{A}BC + B + B\overline{D} + AB\overline{D} + \overline{A}C \\&= C(\overline{A} + A\overline{B}) + B(1 + \overline{D} + A\overline{D}) \\&= C(\overline{A} + A)(\overline{A} + \overline{B}) + B(1) \\&= C\overline{A} + C\overline{B} + B \\&= C\overline{A} + (\overline{B} + B)(B + C) \\&= B + C + C\overline{A} \\&= B + C(1 + \overline{A}) \\&= B + C\end{aligned}$$

Q4. Reduce $(B + BC)(B + B'C)(B + D)$

$$\begin{aligned}&= (B + BC)(B + B'C)(B + D) \\&= (BB + BCB + BB'C + BCB'C)(B + D) \\&= (B + BC + 0 + 0)(B + D) = B(1 + C)(B + D) \\&= B(B + D) \\&= BB + BD = B(1 + D) \\&= B\end{aligned}$$

Problems

REDUCE

$$Q1. A + B[AC + (B + \bar{C})D]$$

$$Q2. (\overline{A + \overline{BC}})(\overline{AB} + ABC)$$

$$Q3. \overline{A}BC + B + B\overline{D} + AB\overline{D} + \overline{A}C = B + C$$

$$Q4. (A + AB)(B + BC)(C + AB)$$